

Chapter 1 - Introduction

INTRODUCTION

1.1 Introduction

Air conditioning is an important part of modern buildings as it helps to maintain suitable indoor temperature, humidity and comfort conditions. It is commonly used in offices, classrooms, laboratories, hospitals, commercial buildings and other occupied spaces. The proper performance of an air-conditioning system depends on selecting equipment according to the actual cooling requirement of the space.

The cooling load of a room is the amount of heat that must be removed from the space to maintain the required indoor conditions. Cooling load calculation is the process of estimating this heat removal requirement by considering the different sources of heat entering or generated within the conditioned space.

The cooling requirement of a room depends on several factors and cannot be determined accurately only from the floor area. Room dimensions, building construction, outdoor temperature, solar radiation, number of occupants, lighting, electrical equipment, ventilation and infiltration all affect the total cooling load.

Heat enters a conditioned space through walls, roof, floor, doors and windows due to temperature differences between indoor and outdoor conditions. Solar radiation can also contribute to heat gain, especially through windows. In addition, occupants, lighting and electrical equipment generate heat inside the room. Outdoor air entering through ventilation and infiltration may further increase the cooling requirement.

An accurate cooling load calculation is therefore necessary for selecting an appropriate air-conditioning capacity. If the system is undersized, it may not maintain the desired indoor conditions. If it is significantly oversized, it may result in unnecessary equipment and operating costs. Hence, cooling load calculation is an important part of HVAC design and evaluation.

For the present project, the cooling load of the HOD cabin is considered. The actual physical characteristics and usage conditions of the cabin will be taken into account to estimate its cooling requirement.

1.2 Need and Importance of Cooling Load Calculation

Different rooms have different cooling requirements because their size, construction, occupancy and operating conditions are not the same. Therefore, a systematic calculation is required to determine the cooling capacity suitable for a particular space.

The main sources contributing to cooling load are:

1. Heat transfer through walls, roof, floor, doors and windows.
2. Solar heat gain through windows and exposed surfaces.
3. Heat generated by occupants inside the room.
4. Heat produced by lighting fixtures.
5. Heat generated by electrical and electronic equipment.
6. Heat entering with ventilation air.
7. Heat entering through infiltration and openings.

Cooling load calculation is important because it:

- Helps determine the required cooling capacity of an air-conditioning system.
- Helps avoid unnecessary oversizing or undersizing of equipment.
- Provides a systematic estimation of different heat gains.
- Assists in selecting suitable HVAC equipment.
- Helps in understanding the thermal behaviour of a conditioned space.
- Provides useful information for evaluating an existing or proposed air-conditioning system.
- Helps identify the major sources of heat gain in a room.

Thus, proper cooling load estimation provides a technical basis for air-conditioning system selection and contributes to efficient HVAC design.

1.3 Factors Affecting Cooling Load

The total cooling load of a conditioned space is influenced by several external and internal factors.

1.3.1 Building Dimensions and Construction

The length, breadth and height of the room determine its floor area, surface areas and volume. The construction and materials of walls, roof, floor, doors and windows affect the rate at which heat is transferred into the space.

1.3.2 Outdoor Temperature and Solar Radiation

The temperature difference between the indoor and outdoor environment affects heat transfer through the building surfaces. Solar radiation is another important source of heat gain, particularly through windows. The amount of solar heat depends on factors such as orientation, time of day, season and characteristics of the window.

1.3.3 Occupants

People inside a room release sensible and latent heat due to their metabolic activity. Therefore, the number of occupants and their level of activity influence the cooling requirement of the space.

1.3.4 Lighting and Electrical Equipment

Lighting fixtures and electrical equipment consume energy and release heat into the room during operation. Computers, printers, fans and other appliances can therefore contribute to the internal cooling load.

1.3.5 Ventilation and Infiltration

Ventilation introduces outdoor air into the conditioned space, while infiltration occurs when outdoor air enters through doors, windows and other openings. If the entering air has a higher temperature or humidity than the indoor air, additional cooling is required.

Therefore, actual information regarding the room dimensions, openings, occupancy, lighting and equipment is important for obtaining a realistic cooling load calculation.

1.4 Aim and Objectives of the Project

Aim

To calculate the cooling load of the HOD cabin by considering the different sources of heat gain and determine the cooling requirement of the selected space.

Objectives

The main objectives of the project are:

1. To study the basic principles and methods of cooling load calculation.
2. To measure and record the actual dimensions and physical details of the HOD cabin.
3. To identify the different sources of heat gain affecting the selected space.
4. To consider heat transfer through walls, roof, floor, doors and windows.
5. To estimate the heat gain due to occupants, lighting and electrical equipment.
6. To consider the effect of ventilation and infiltration through doors, windows and other openings.
7. To calculate the individual components of cooling load using suitable engineering methods.
8. To determine the total cooling load required for the HOD cabin.
9. To analyse the contribution of different heat sources to the overall cooling requirement.
10. To provide a technical basis for understanding the air-conditioning requirement of the selected space.